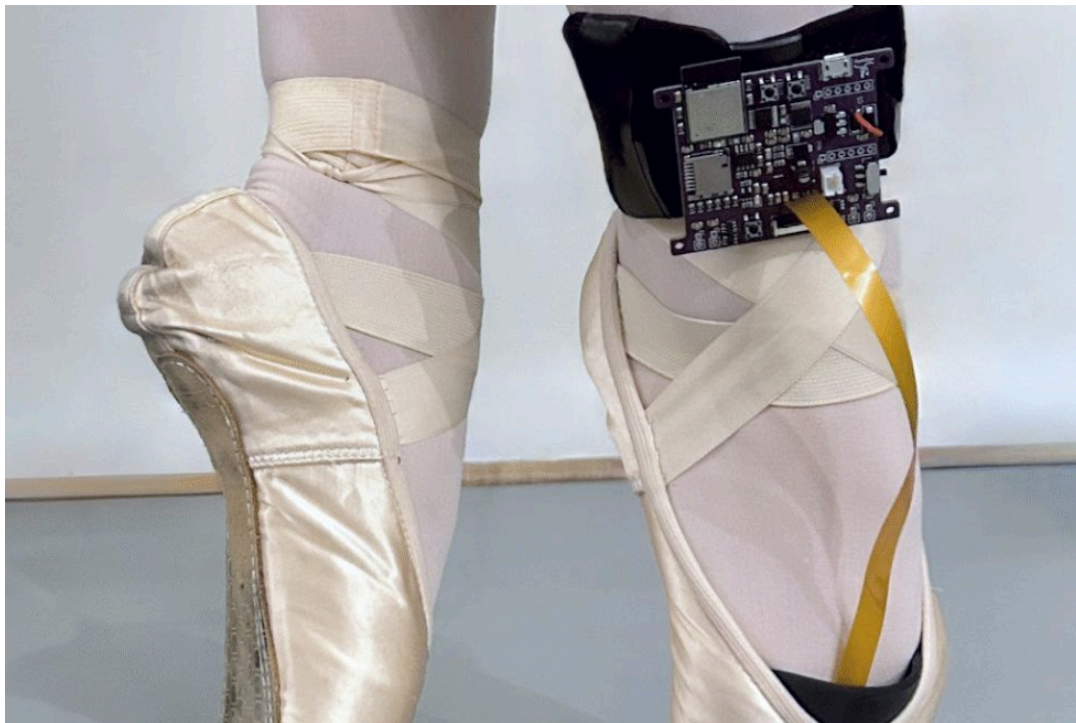


1. **“Engineering Meets Ballet: A Smart Solution for Safer Pointe Work”** - this article demonstrates an example of a system wherein a wearable device is created to analyse the pressure distribution using 64 pressure sensors in the front part of the dancers foot and send signals to the dancer through an ankle band that provides feedback via haptics (vibration), since it provides intuitive aid while a dancer is practicing. The sensors are inserted into the shoe in the form of a toe pad (like a sock which fits into the shoe) and the sensors used are flexible to accommodate this. Integrating this into the shoe in this way prevents the sensors from being an interference to the dancer. The team pursuing this product faced the following challenges in the process:

- They need a diverse database of dancers with different foot flexibility, strength and shapes to test their product
- Since dancers have personalised needs, making a product that is generalised has been a challenge for them
- Making the system untethered, lightweight and unobtrusive is challenging due to space constraints in the pointe shoe and keeping in mind the mobility of the dancer



Citation:

<https://engineering.jhu.edu/news/engineering-meets-ballet-a-smart-solution-for-safer-pointe-work/>

2. **“Biomechanical Comparison of ‘Dead’ and ‘New’ Pointe Shoes in Female Professional Ballet Dancers (2019)”** - This paper studied a comparison between new pointe shoes that had 5-50 hours of training and dead pointe shoes that had 51-400 hours. The movements tested on dancers were relevé à la second, sous-sus and a single pirouette. By measuring the centre of pressure sway area using a force plate and Ground Reaction Forces (GRF) in three directions: anterior-posterior, medial lateral and vertical the following was found:

- “Dead” shoes showed a significantly larger ‘sway area’ from the mean centre of pressure, impacting stability
- No statistically significant differences in force directions (vertical, medial lateral) across the two shoe conditions
- ‘Dead’ shoes provided less stability and reduced balance whereas ‘new’ shoes allowed more force production in certain movements

Citation:

<https://thesportjournal.org/article/biomechanical-comparison-of-dead-and-new-pointe-shoes-in-female-professional-ballet-dancers/>

3. **“A Wearable Insole System to Measure Plantar Pressure and Shear Stress”** - this paper details the design, development and testing of a wearable insole system capable of measuring plantar pressure and shear stress under the foot in real time, for diabetic foot ulcer prevention. The system utilises thin, flexible TRIPS sensors placed at the heel, 1st and 5th metatarsals and the hallux. The sensors recorded pressure and shear in two orthogonal directions and are integrated into an insole. The final product showed high accuracy demonstrating that the multi-sensor approach and flexible insole design is feasible for monitoring pressure distribution in pointe shoes.

Citation: <https://doi.org/10.3390/s23063126>

4. **“The Pressure Pointe (2025)”** - this project aimed to reduce high pressure on a dancers foot in pointe shoes, by specifically improving upon the pressure distribution in the tow box of the pointe shoes. The project focused on different materials inserted into the box to find which one was most

effective in redistributing pressure among the toes, especially reducing stress on the big toe. A model foot was used to test the product, rather than a human. Several prototype inserts were created made of materials like spray foam, soft foam, rigid plastic, memory foam etc. They used pressure indicating film to identify the pressure distribution. Through conducting this experiment, the following was found:

- Among all materials tested, spray foam performed the best as it had the lowest average percentage area of toe that was unloaded. This means most of the pressure was distributed equally amongst all five toes.
- The second best was soft foam and the worst material was rigid plastic
- Spray foam was also liked by consumers as they found it improves support and comfort

Citation: <https://www.sensorprod.com/research-articles/the-pressure-pointe/>

5. **“Sensing Technology for Assessing Motor Behaviour in Ballet (2022)”** - This paper provides a comprehensive summary of sensing technologies used to assess motor behaviour in ballet. After examining various studies, it was concluded that force plates, electromyography (EMG) and inertial sensors are the most frequently used methods for evaluating classical ballet movements. Certain important findings from this paper are:

- 55% of studies used two or more types of technology, most commonly combining kinematic (motion capture) and kinetic (force plate) analyses for better data
- Only 29 distinct ballet movements have been analysed and studied in detail with sauté, grande jeté and pas de chat being studied the most
- EMG is used to analyse muscle activation in movements like pliés

Citation: <https://doi.org/10.1186/s40798-022-00429-8>.

6. **“A Smart Wearable Device for Monitoring and Self-Assessment in Sports”** - This study presents a smart wearable device for sports and rehabilitation applications that can monitor and self-assess movement using multi-sensor technology. It integrates pressure sensors, temperature sensors, and moisture sensors within a shoe to continuously collect and wirelessly transmit data. These sensors accurately map plantar pressure and environmental parameters during activity and send data wirelessly to a paired app or receiver for analysis and feedback. Results show it reliably tracks weight distribution, detects changes in gait and loading, and contributes valuable feedback for athlete training, rehabilitation, and injury prevention.

Citation: <https://doi.org/10.1016/j.ijmedinf.2020.104343>

7. **“Pressure Measurement on Toes in Pointe Shoes”** - This study introduces a novel pressure measurement system using thin, flexible capacitive sensors embedded in pointe shoes to quantify toe pressure distribution during ballet movements. The sensors, placed across the toe box, captured real-time data on pressure hotspots in professional dancers performing relevés and jumps, revealing peak loads exceeding 200% body weight on the first metatarsophalangeal joint. Key findings include significant pressure asymmetry between toes, with the hallux bearing 40-50% of total load, and correlations between sensor data and self-reported foot pain.

Challenges addressed in the study:

- Sensor flexibility to avoid altering shoe dynamics
- Wireless data transmission for untethered performance
- Calibration for varying foot shapes

Citation: <https://journals.sagepub.com/doi/10.1177/1089313X251330462>

8. **“Wearable Accelerometer for Ballet Jump Analysis”** - This paper evaluates a shoe-mounted accelerometer and gyroscope system for assessing ballet jump height, landing forces, and aerial time in variations like sautés and jetés. Worn by 12 dancers, the IMU sensors provided 95% accuracy against force plates, identifying technique flaws linked to sesamoiditis from harsh landings. The untethered setup enabled field testing without lab constraints.

Key findings:

- Jump heights averaged 0.4m with 2-3x body weight impacts
- Real time feedback improved landing symmetry

Citation: <https://pubmed.ncbi.nlm.nih.gov/31718476/>

9. **“Pressure Distribution and Foot Injury in Ballet Dancers”** - The research used force-sensitive insoles in pointe shoes to analyze pressure patterns across 20 professional ballerinas during pliés, relevés, and pointe work. Results showed metatarsal heads endured 1.5-2.5x normal loads, with rigid shanks exacerbating hallux rigidus risks; flexible toe pads mitigated peaks by 25%

Relevant challenges:

- Individual foot morphologies required custom sensor arrays
- Long term wear caused sensor drift (this is something to take into consideration while building our prototype)

Citation: <https://pubmed.ncbi.nlm.nih.gov/22687718/>

10. “Measuring Foot Pressure in Pointe Shoes of Ballet Dancers” - Researchers developed a wearable insole with 16 FSR sensors integrated into pointe shoes to map plantar pressure during dynamic ballet tasks like pirouettes and grand battements. The system logged data at 100 Hz, showing "dead" shoes increased peak pressures by 15-20% under the metatarsals compared to new ones, heightening injury risk. Haptic feedback prototypes reduced overload incidents by 30% in trials with elite dancers.

Notable outcomes:

- Higher sway in centre of pressure with worn shoes
- Need for dancer specific calibration
- Lightweight design under 20g per shoe

Citation: <https://pubmed.ncbi.nlm.nih.gov/40098421/>

